VISVESVARAYATECHNOLOGICALUNIVERSITY
BELAGAVI



Project Report on

"BRAIN TUMOR DETECTION USING CNN"

Submitted in the partial fulfillment for the requirements of the degree of

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING

Submitted By

BOGATHI ANUDEEPA	**1BY19CS029**
DEEKSHITH KORIPALLI	**1BY19CS040**
KODIDALA BHAVITHA	**1BY19CS068**
KURLI KAILASH REDDY	**1BY19CS069**

Under the guidance of

MR ANAND R

ASSISTANT PROFESSOR
Department of CSE,
BMSIT&M



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT
YELAHANKA, BENGALURU - 560064.

2022-2023

VISVESVARAYATECHNOLOGICALUNIVERSITY
BELAGAVI

BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT
YELAHANKA, BENGALURU – 560064

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Project work entitled **"BRAIN TUMOR DETECTION USING CNN"** is a bonafide work carried out by **BOGATHI ANUDEEPA (1BY19CS029), DEEKSHITH KORIPALLI (1BY19CS040), KODIDALA BHAVITHA (1BY19CS068), KURLI KAILASH REDDY (1BY19CS069),** in partial fulfillment for the award of **Bachelor of Engineering Degree in Computer Science and Engineering** of the **Visvesvaraya Technological University, Belagavi** during the year 2022-2023. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in this report. The project report has been approved as it satisfies the academic requirements in respect of project work for B.E. Degree.

Signature of the Guide	**Signature of the HOD**	**Signature of Principal**
Mr. Anand. R Assistant Professor Dept. of CSE, BMSIT&M	Dr. Thippeswamy G Professor & HOD, Dept. of CSE, BMSIT&M	Dr. Mohan Babu G N Principal, BMSIT&M

External VIVA-VOCE

Name of the Examiners	Signature with Date
1.	
2.	

ACKNOWLEDGEMENT

We are happy to present this project after completing it successfully. This project would not have been possible without the guidance, assistance and suggestions of many individuals. We would like to express our deep sense of gratitude and indebtedness to each and every one who has helped us make this project a success.

We heartily thank our Principal**, Dr. MOHAN BABU G N, BMS Institute of Technology and Management,** for his constant encouragement and inspiration in taking up this project.

We heartily thank our Head of the **Department, Dr. Thippeswamy G, Department of Computer Science and Engineering, BMS Institute of Technology and Management,** for his constant encouragement and inspiration in taking up this project.

We gracefully thank our Project Guide, Mr Anand R, **Assistant Professor, Department of Computer Science and Engineering,** for his guidance, support, and advice.

Special thanks to all the teaching and non-teaching staff members of Computer Science and Engineering Department for their help and kind co-operation.

Lastly we thank our parents and friends for the support and encouragement given to us in completing this precious work successfully.

BOGATHI ANUDEEPA (1BY19CS029)
DEEKSHITH KORIPALLI (1BY19CS040)
KODIDALA BHAVITHA (1BY19CS068)
KURLI KAILASH REDDY (1BY19CS069)

DECLARATION

We, **Bogathi Anudeepa (1BY19CS029), Deekshith Koripalli (1BY19CS040), Kodidala Bhavitha (1BY19CS068), Kurli Kailash Reddy (1BY1CS069),** students of Eight semester B.E, in the Department of Computer Science and Engineering, BMS Institute of Technology and Management, Bengaluru declare that the project work entitled **"Brain Tumour Detection Using CNN"** has been carried out by us and submitted in partial fulfillment of the course requirements for the award of degree in Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2022- 2023. The matter embodied in this report has not been submitted to any other universityor institution for the award of any other degree or diploma.

Bogathi Anudeepa	1BY19CS029
Deekshith Koripalli	1BY19CS040
Kodidala Bhavitha	1BY19CS068
Kurli Kailash Reddy	1BY19CS069

Department of Computer Science and
Engineering, BMS Institute of Technology
and Management, Bengaluru
-560064.

ABSTRACT

Brain Tumor segmentation is one of the most crucial tasks in the terrain of medical image processing as a human-assisted manual classification can result in inaccurate prediction and diagnosis. Moreover, it is an aggravating task when there is a large amount of data present to be assisted. Brain tumors have high diversity in appearance and there is a similarity between tumor and normal tissues and thus the extraction of tumor regions from images becomes unyielding. In this design, we proposed a method to extract brain tumor from 2D Magnetic Resonance brain Images (MRI) by Fuzzy C-Means clustering algorithm which was followed by traditional classifiers and convolutional neural network. The experimental study was carried on a real-time dataset with diverse tumor sizes, locations, shapes, and different image intensities. In the traditional classifier part, we applied Support Vector Machine (SVM),This design can not only predict the tumor but also it can classify the type of tumor along with its symptoms and accuracy of it's prediction.

ACKNOWLEDEMENT **I**
DECLARATION **II**
ABSTRACT **III**
TABLE OF CONTENTS **IV**
LIST OF FIGURES **VI**

TABLE OF CONTENTS

CHAPTER NO	TOPIC NAME	PAGE NO
1.	**INTRODUCTION**	**1**
	1.1.Background	**3**
	1.2. Motivation	**4**
	1.3. Problem Statement	**5**
	1.4. Objectives	**6**
	1.5. Scope	**6**
	1.6. Challenges	**7**
	1.7. Organization of the thesis	**8**
2.	**LITERATURE SURVEY**	**10**
3.	**REQUIREMENTS**	**18**
	3.1 Non-Functional Requirements	**19**
	3.2 System Requirements	**20**
	3.2.1 Hardware Requirements	**20**
	3.2.2 Software Requirements	**20**
4.	**DETAILED DESIGN/ARCHITECTURE**	**21**
	4.1 System Architecture	**21**
	4.2 Flow Chart	**22**
	4.3 Use Case Diagram	**23**
	4.4 Data Flow Diagram	**23**
	4.4.1 Level 0 DFD	**23**
	4.4.2 Level 1 DFD	**24**
	4.42 Level 2 DFD	**25**

5.	**IMPLEMENTATION**	**26**
	5.1 SDLC(System Development Life Cycle)	**26**
	5.2 Methodology	**26**
	5.2.1 Planning	**27**
	5.2.2 Data Collection	**27**
	5.2.3 Data Pre-Processing	**28**
	5.2.4 Analysis	**30**
6.	**TESTING**	**33**
	6.1 CNN	**34**
	6.1.1 Confusion Matrix	**34**
	6.1.2 Classification for CNN	**35**
7.	**RESULT**	**36**
	7.1 Convolution Neural Network	**36**
	7.1.1 CLSTM-Report On Classification	**36**
	7.1.2 CNN-Confusion Matrix	**37**
8.	**CONCLUSION**	**42**
	8.1 Future Scope	**43**

List of Figures

S.no	FIGURE NAME	PAGE NO
4.1	System Architecture	**21**
4.2	Flow Chart	**22**
4.3	Use Case Diagram	**23**
4.1.1	Level 0 DFD	**24**
4.1.2	Level 1 DFD	**24**
4.1.3	Level 2 DFD	**25**
5.1	Software Development Life Cycle	**26**
6.1	Testing Files	**33**
6.1.1	Confusion Matrix for CNN and Formulas	**34**
6.1.2	Confusion Matrix for CLSTM	**34**
6.1.3	Classification Report for CNN	**35**
7.1	Confusion Matrix for CLSTM	**36**
7.2	Confusion Matrix for CLSTM	**37**
7.2.1	Login Page	**38**
7.2.2	Home Page	**38**
7.2.3	Changing Password	**39**
7.2.4	Detection of Tumor (Glimoa)	**39**
7.2.5	Detection of Tumor (Meningioma)	**40**
7.2.7	Detection of Tumor (No Tumor)	**41**
7.2.8	Detection of Tumor (Pitutary)	**41**

CHAPTER 1

INTRODUCTION

Medical image is the main carrier and important carrier of modern medical alternative language description. It gradually plays an increasingly important role in the medical field with very nutritive expression and objective description. Medical image analysis is a modern image analysis technology that integrates mathematical modeling, artificial intelligence, integrated medical imaging, digital image processing and other multidisciplinary.

Denting whether a tumor is benign or malignant according to the image is also a subject that the medical community has not stopped. The feature extraction of the image is to first segment the image, and then extract the texture features in the image after segmentation. There are a lot of textures on trees. These things that can be seen visually are taken from their textures. Of course, medically, for tumors, it is not as intuitive as trees or clothes. It can only be sliced into tumors, relying on imaging equipment to make its texture features into medical images such as CT or MRI. Texture is a basic and very useful information feature in an image. It is an important parameter for describing image content, and its academic research has gradually become a new topic.

Magnetic resonance imaging (MRI) provides detailed information about brain tumor anatomy, cellular structure and vascular supply, making it an important tool for the effective diagnosis, treatment and monitoring of the disease. Magnetic resonance imaging (MRI) is a non-invasive medical test that helps physicians diagnose and treat medical conditions. MRI uses a powerful magnetic field, radio frequency pulses and a computer to produce detailed pictures of organs, soft tissues, bone and virtually all other internal body structures. The images can then be examined on a computer monitor, transmitted electronically, printed or copied to a CD. MRI does not use ionizing radiation (x-rays). Detailed MR images allow physicians to evaluate various parts of the body and determine the presence of certain diseases. Automated brain tumor detection from MRI images is one of the most challenging task in today's modern Medical imaging research. Automatic detection requires brain image segmentation, which is the process of partitioning the image into distinct regions, is one of the most important and challenging aspect of computer aided clinical diagnostic tools. Noises present in the Brain MRI images are multiplicative noises and reductions of these noises are difficult task. The minute anatomical details should not be destroyed by the process of noise removal from clinical point

of view. These makes accurate segmentation of brain images a challenge. However, accurate segmentation of the MRI images is very important and crucial for the exact diagnosis by computer aided clinical tools.

A large variety of algorithms for segmentation of MRI images had been developed. Surgical planning, postsurgical assessment, abnormality detection, and much other medical application require medical image segmentation. In spite of wide number of automatic and semi – automatic image segmentation techniques, they fail in most cases largely because of unknown and irregular noise, in-homogeneity, poor contrast and weak boundaries which are inherent to medical images.

A tumor inside the brain is an irregular development of a tissue that can cause problems with brain function. Doctors classify tumors based on where the tumor cells arose fromand whether or not they are cancerous. Benign tumors may grow and influence the remaining healthy brain tissue. Malignant tumors are generally called brain cancer that grows outside the brain. In the United States currently, an approximate 700,000 people have a primary brain tumor, with an estimated 85,000 predicted to be diagnosed by 2021. Brain tumors can be lethal, have a significant impact on people's lives, and fully change the lives of both the patient and their families. They effect male, female, and infants of all cultures and nationalities, without regard for race or ethnicity. In brain tumor diagnosis and treatment, imaging technology plays a key role. Imaging of tumors can be done in many different forms, including CT scanning, MRI, etc.

A Brain MR image contains an enormous volume of information about the spatial brain structure and is suitable for medical diagnosis. Tumors of the brain are considered one of the dangerous and most complex for diagnosis and treatment. It can be seen that different methods are used to identify brain tumors from MR images. A CNN is an artificial neural network which is specifically designed to undertake processing these are strong image processing and artificial intelligence systems that use deep learning to perform both generative and descriptiive functions,often involving machine vision, such as image and video recognition, as well as recommender systems and natural language processing.

1.1 Background

Convolutional Neural Networks (CNNs): CNNs are deep learning models specifically designed for analyzing visual data such as images. They are composed of multiple layers, including convolutional layers, pooling layers, and fully connected layers. CNNs learn to automatically extract relevant features from input images and make predictions based on those features.

Brain Tumor Imaging: Magnetic Resonance Imaging (MRI) is commonly used to capture detailed images of the brain. MRI scans provide high-resolution images that can reveal abnormalities, including brain tumors. These scans are typically represented as a collection of 2D or 3D images called slices.

Data Preprocessing: Prior to feeding the MRI scans into a CNN model, data preprocessing is essential. This step involves normalizing the image intensities, resizing the images to a uniform size, and sometimes applying additional transformations such as cropping or augmenting the data to increase the size of the training dataset.

Training Data: A CNN model for brain tumor detection requires a large and diverse dataset of brain MRI scans. This dataset should include both tumor and non-tumor examples. Each scan is labeled accordingly to indicate the presence or absence of a tumor.

Model Training: The CNN model is trained using the labeled MRI dataset. During training, the model learns to extract relevant features from the input scans through the convolutional layers and then classify those features as tumor or non-tumor using the fully connected layers. The model is optimized by adjusting the weights and biases based on the computed loss and the desired output.

Model Evaluation: Once the model is trained, it is evaluated using a separate dataset that was not used during training (i.e., a validation or test set). The evaluation metrics commonly used for brain tumor detection include accuracy, sensitivity (recall), specificity, and the area under the receiver operating characteristic (ROC) curve.

Prediction and Interpretation: After the model is trained and evaluated, it can be used to predict the presence of a brain tumor in new, unseen MRI scans. The model examines the image and outputs a probability or a binary prediction indicating the likelihood of a tumor presence.

1.2 Motivation

The motivation behind a “Brain Tumor Detection using CNN (Convolutional Neural Network)” project is to develop an accurate and efficient method for early detection and diagnosis of brain tumors. Brain tumors are abnormal growths of cells in the brain that can cause various neurological symptoms and potentially be life-threatening. Detecting brain tumors at an early stage is crucial for providing timely treatment and improving patient outcomes.

Accuracy: CNNs have demonstrated exceptional performance in various image recognition tasks, including medical imaging. By leveraging their ability to learn complex patterns and features from images, CNNs can potentially achieve high accuracy in detecting brain tumors.

Efficiency: Traditional methods of brain tumor detection often require manual interpretation of medical images by radiologists, which can be time-consuming and subject to human error. CNN-based approaches can automate the detection process, reducing the time and effort required for analysis and improving efficiency.

Early diagnosis: Early detection of brain tumors can significantly impact treatment options and patient outcomes. By using CNNs to detect tumors at an early stage, healthcare professionals can initiate appropriate interventions promptly, potentially leading to better prognosis and survival rates.

Non-invasive approach: Brain tumor detection using CNNs relies on non-invasive medical imaging techniques, such as magnetic resonance imaging (MRI) or computed tomography (CT) scans. This eliminates the need for invasive procedures, such as biopsies, which can be uncomfortable for patients and carry additional risks.

Scalability: CNN-based models can be trained on large datasets, allowing for scalability and generalization to different types of brain tumors. This scalability enables the development of robust and versatile brain tumor detection systems that can be applied in various healthcare settings.

1.3 Problem Statement

The objective of this project is to develop a deep learning model based on Convolutional Neural Networks (CNN) for accurate and efficient brain tumor detection using medical imaging data. The problem at hand is the timely and accurate identification of brain tumors in medical images, which is crucial for early diagnosis and treatment planning.

Current methods for brain tumor detection rely heavily on manual analysis by radiologists, which is time-consuming and prone to human error. Furthermore, the increasing volume of medical imaging data makes it challenging to process and analyze images manually. Therefore, there is a need for an automated system that can accurately and efficiently identify brain tumors in medical images.

Brain tumors are a significant health problem worldwide, with an estimated incidence of over 250,000 new cases each year worldwide. These tumors can be benign (non-cancerous) or malignant (cancerous), and can occur at any age.

The diagnosis and treatment of brain tumors can be challenging due to the complex anatomy of the brain and the potential for significant morbidity and mortality. Current diagnostic approaches often involve a combination of imaging techniques such as CT scans and MRI, as well as biopsy and pathological analysis of tissue samples. Treatment options may include surgery, radiation therapy, and chemotherapy, depending on the type and location of the tumor, as well as the age and overall health of the patient. Still there are no quality diagnostic techniques for brain tumors to identifying the type of the tumor.

1.4 Objectives

The aim of tumor detection and classification using image processing is to accurately identify and classify different types of tumors in medical images, such as MRI and CT scans. This can help in early diagnosis and treatment of various types of cancers.

The following criteria are listed dependent on the purpose with this survey:

- Developing image processing algorithms that can accurately detect and segment tumors from medical images.
- Developing machine learning models that can classify different types of tumors based on their characteristics.
- Evaluating the performance of the developed algorithms and models using various metrics such as sensitivity, specificity, and accuracy.
- Predicting the probability percentage of the brain tumor in the given MRI scan.
- Displaying the symptoms for the type of tumor predicted by the design.
- Decreasing the time required for manual process such as conducting biopsy, and also decreasing the chances of human errors in diagnosis.

1.5 Scope

The scope of a brain tumor detection project using convolutional neural networks (CNN) and machine learning (ML) techniques can be quite significant, some key aspects that could be covered within the project's scope are

Hyperparameter Tuning: Conducting experiments to optimize the hyperparameters of the CNN model. These hyperparameters include learning rate, batch size, number of layers, and filter sizes. Fine-tuning these parameters can improve the model's performance.

Deployment and User Interface: Developing a user-friendly interface where medical professionals can input brain images for tumor detection. The deployed model should be able to provide real-time predictions on new input images.

Ethical Considerations: Considering the ethical implications of the project, including privacy concerns, data security, and ensuring proper consent and compliance with relevant regulations and standards.

Performance Comparison: Comparing the performance of the CNN-based approach with other ML techniques or traditional methods used for brain tumor detection, such as manual analysis by radiologists. This can demonstrate the potential advantages of the proposed CNN-ML model.

1.6 Challenges

- **Limited and Imbalanced Data:** Obtaining a large and diverse dataset of brain tumor images can be challenging. Additionally, the distribution of tumor and non-tumor images may be imbalanced, with fewer examples of tumor images. This can impact the model's ability to learn and generalize effectively.
- **Image Quality and Variability:** Brain tumor images can vary in terms of resolution, quality, and imaging techniques (such as MRI or CT scans). Variations in brightness, contrast, and image artifacts can make it difficult to extract meaningful features and patterns from the images, affecting the accuracy of the model.
- **Localization and Segmentation:** Identifying the precise location and boundaries of a tumor within an image is crucial for accurate detection. This requires advanced image segmentation techniques to separate the tumor region from surrounding brain tissues. Achieving accurate tumor segmentation can be challenging, particularly when dealing with irregular tumor shapes, overlapping structures, or heterogeneous tumor appearances.
- **Interpretability and Explainability:** CNNs are complex models with a large number of parameters. Understanding how the model arrives at its predictions and providing explanations for its decisions is important in medical applications. Interpreting the inner workings of CNNs and providing meaningful insights to clinicians is a challenge, as CNNs are often treated as black-box models.
- **Generalization to Unseen Cases:** CNNs trained on a specific dataset may not generalize well to unseen brain tumor images. This is particularly true if the testing data significantly differs from the training data in terms of imaging protocols, patient demographics, or tumor characteristics. Ensuring the model's ability to generalize to diverse cases is an ongoing challenge.
- **Computational Complexity:** Training deep CNN models for brain tumor detection requires substantial computational resources and time. Processing high-resolution medical images can be computationally intensive, making it challenging to train models

efficiently. Deploying these models on resource-constrained devices or in real-time applications also poses practical challenges.

1.7 Organization of the Thesis

The organization of the thesis Brain Tumor Detection project involves data collection, preprocessing, image processing, CNN design, training, validation, testing, deployment, and continuous evaluation and improvement to develop an accurate and efficient system for brain tumor detection.

1. **Data Collection**: Gather a comprehensive dataset of brain tumor images, including both positive (images with tumors) and negative (healthy brain images) samples. These images can be obtained from medical databases, hospitals, or research institutions.

2. **Preprocessing:** Clean and preprocess the collected images to ensure they are in a suitable format and resolution for further analysis. Preprocessing steps may include resizing, normalization, noise removal, and contrast enhancement to improve the quality of the images.

3. **Image Processing:** Utilize image processing techniques to extract relevant features from the preprocessed images. This may involve filtering, edge detection, morphological operations, and other image enhancement methods to highlight tumor regions and suppress noise.

4. **Data Augmentation:** Enhance the dataset by applying various data augmentation techniques such as rotation, scaling, flipping, and adding noise. Data augmentation helps to increase the diversity of the dataset and improve the robustness of the model.

5. **Convolutional Neural Network (CNN) Design:** Design and implement a CNN architecture suitable for brain tumor detection. A CNN is a deep learning model that is capable of automatically learning and extracting features from images. The architecture may consist of multiple convolutional layers, pooling layers, and fully connected layers.

6. **Training:** Train the CNN using the preprocessed and augmented dataset. This involves feeding the images into the network and adjusting the model's parameters to minimize the difference between predicted and ground truth tumor labels. Training is typically performed using gradient descent optimization algorithms such as stochastic gradient descent (SGD) or Adam.

7. **Validation:** Evaluate the performance of the trained model using a separate validation dataset. This step helps to assess the model's generalization ability and detect any overfitting issues. Various metrics such as accuracy, precision, recall, and F1 score can be calculated to measure the model's performance.

8. **Testing:** Test the trained model on a separate test dataset that was not used during training or validation. This step provides an unbiased evaluation of the model's performance and allows for comparisons with other existing methods or models.

9. **Deployment:** Once the model has been validated and tested, it can be deployed in a real-world setting. This may involve integrating the model into a user-friendly interface or application that can accept input brain images and provide tumor detection results.

10. **Evaluation and Improvement:** Continuously monitor and evaluate the performance of the deployed system, collect user feedback, and make improvements as necessary. This can involve retraining the model with new data, optimizing hyperparameters, or updating the system based on new advancements in brain tumor detection research.

CHAPTER 2

LITERATURE SURVEY

- S. A. Al-Maadee d et al. (2021), “3D CNNs for brain tumultuous segmentation”, The study reviews recent developments in 3D CNN-based methods for brain tumor segmentation in MRI images. The review covers various aspects such as network architectures, training strategies, and evaluation metrics.
- A. S. Alshamma ri et al. (2021), "A comparison of deep learning algorithm algorithms for brain tumor segmentation on MRI images: a systematic view, The study compares the performance of different deep learning algorithms for brain tumour segmentation in MRI images..
- S. El Asri et al. (2021), "Convolut ional neural networks for brain tumor segmentation”, The study provides a systematic review of CNN-based methods for brain tumor segmentation in MRI images.
- N. Singh et al. (2021), "Brain tumor classifica tion using transfer learning with deep convoluti onal neural networks ", The authors fine-tuned a pre-trained CNN model (VGG-16) on a dataset of 306 MRI images (154 tumor and 152 non-tumor) and evaluated its performance on a test dataset of 54 MRI images.
- S. R. Azad et al. (2021), "Deep LearningBased Automati c Detection of Brain Tumor Using Magnetic Resonan ce Images", The proposed method used a modified VGG-16 CNN architecture, which was trained on a dataset of 306 brain MRI images, including 154 images with tumors and 152 images without tumors.
- A. R. Tariq et al. (2019), "Brain Tumor Detection and Classification Using Convoluti onal Neural Network", The proposed method used a modified AlexNet CNN architecture, which was trained on a dataset of 253 brain MRI images, including 98 images with glioma tumors, 76 images with meningioma tumors, and 79 images with no tumors.
- M. M. Kamble and M. M. Kodabagi (2022), "Convolut ional neural networks for brain tumor segment ation”, The review covers various aspects such as preprocessing techniques, network architectures, and training strategies.
- H. Ma et al. (2021), "Brain tumor segmentation using 3D convolutional neural networks with dilated convoluti ons"”, The authors used a modified U-Net architecture with dilated convolutions to capture more context information in 3D images.

- Y. Wu et al.(2021),"Deep learning for glioma segmentation",The study review recent developments in deep learning-based methods for glioma segmentation. The review covers various aspects such as network architectures,training strategies,and evaluation metrics.
- S. S. Chennu et al. (2021) "Automat ed detection of meningio ma using convoluti onal neural networks on multi-seq uence MRI" The study proposes a convolutional neural network-based method for automated detection of meningioma using multi-sequence MRI. The proposed method includes pre-processing, feature extraction, and classification.
- Y. Zhou et al. (2021), "Multimodal brain tumor segmentation using cascaded 3D CNNs with model aggregation and label fusion", The study proposes a cascaded 3D convolutional neural network-based method for multimodal brain tumor segmentation. The proposed method includes cascaded segmentation, model aggregation, and label fusion.
- H. Qi et al. (2021), "Glioblast oma multiform e detection using deep learning", The study reviews recent developments in deep learning-based methods for glioblastoma multiforme detection. The review covers various aspects such as network architectures, training strategies, and evaluation metrics.
- M. Shamsi et al. (2021), "A review of machine learning technique s for meningio ma detection " reviews recent developments in machine learning-based methods for meningioma detection. The review covers various aspects such as feature extraction, classification algorithms, and evaluation metrics.
- M. Abadi et al. (2021), "A deep learningbased approach for automati c detection of pituitary adenoma ", a deep learning-based method for automatic detection of pituitary adenoma on MRI. The method involves training a convolutional neural network on a large dataset of MRI images.
- S. Z. Azmi et al. (2021), "A review of machine learning technique s for medullob lastoma detection ", The study reviews recent developments in machine learning-based methods for medulloblastoma detection. The review covers various aspects such as feature extraction, classification algorithms, and evaluation metric

SL nO.	Author and Year	Title	Methodology	Advantages	Limitations
1.	S. A. Al-Maadeed et al. (2021)	“3D CNNs for brain tumor segmentation”	The study reviews recent developments in 3D CNN-based methods for brain tumor segmentation in MRI images. The review covers various aspects such as network architectures, training strategies, and evaluation metrics.	The study provides a comprehensive overview of the recent advances in 3D CNN-based methods for brain tumor segmentation, including network architectures, training strategies, and evaluation metrics.	The study does not provide a direct comparison of different methods and their performance on different datasets.
2.	A. S. Alshammari et al. (2021)	"A comparison of deep learning algorithms for brain tumor segmentation on MRI images: a systematic review"	The study compares the performance of different deep learning algorithms for brain tumor segmentation in MRI images.	It provides a comprehensive comparison of the performance of different deep learning algorithms for brain tumor segmentation.	It only focuses on segmentation and does not cover other tasks such as classification or detection.
3.	S. El Asri et al. (2021)	"Convolutional neural networks for brain tumor segmentation”	The study provides a systematic review of CNN-based methods for brain tumor segmentation in MRI images.	The study provides a comprehensive overview of CNN-based methods for brain tumor segmentation.	The study only focuses on segmentation and does not cover other tasks such as

				The authors identify the challenges and limitations of these methods and provide recommendations for future research.	classification or detection.
4.	N. Singh et al. (2021)	"Brain tumor classification using transfer learning with deep convolutional neural networks"	The authors fine-tuned a pre-trained CNN model (VGG-16) on a dataset of 306 MRI images (154 tumor and 152 non-tumor) and evaluated its performance on a test dataset of 54 MRI images.	The proposed method achieved high accuracy in brain tumor classification using MRI images. The use of transfer learning from a pre-trained model reduces the computational cost and training time.	The dataset used in the study is relatively small, and the model's performance on larger datasets remains to be seen.
5.	S. R. Azad et al. (2021)	"Deep Learning-Based Automatic Detection of Brain Tumor Using Magnetic Resonance Images"	The proposed method used a modified VGG-16 CNN architecture, which was trained on a dataset of 306 brain MRI images, including 154 images with tumors and 152 images without tumors.	The proposed method achieved high accuracy in brain tumor detection using MRI images, which can potentially reduce the burden on radiologists.	The dataset used in the study is relatively small, and the model's performance on larger datasets remains to be seen. The study did not include a comparison with other state-of-the-art methods.

6.	A. R. Tariq et al. (2019)	"Brain Tumor Detection and Classification Using Convolutional Neural Network"	The proposed method used a modified AlexNet CNN architecture, which was trained on a dataset of 253 brain MRI images, including 98 images with glioma tumors, 76 images with meningioma tumors, and 79 images with no tumors.	The use of a modified AlexNet architecture makes the model easily reproducible and adaptable to other datasets.	The dataset used in the study is relatively small, and the model's performance on larger datasets remains to be seen. The study did not include a comparison with other state-of-the-art methods.
7.	M. M. Kamble and M. M. Kodabagi (2022)	"Convolutional neural networks for brain tumor segmentation"	The review covers various aspects such as preprocessing techniques, network architectures, and training strategies.	The study provides a comprehensive overview of the recent advances in CNN-based methods for brain tumor segmentation, including preprocessing techniques, network architectures, and training strategies.	he study does not provide a direct comparison of different methods and their performance on different datasets.
8.	H. Ma et al. (2021)	"Brain tumor segmentation using 3D convolutional	The authors used a modified U-Net architecture with dilated convolutions to capture more context	The use of dilated convolutions enhances the network's ability to capture context	The dataset used in the study is relatively small, and the model's performanc

		neural networks with dilated convolutions"	information in 3D images.	information.	e on larger datasets remains to be seen.
9.	Y. Wu et al.(2021)	"Deep learning for glioma segmentation"	The study reviews recent developments in deep learning-based methods for glioma segmentation. The review covers various aspects such as network architectures, training strategies, and evaluation metrics.	It provides a comprehensive overview of the recent advances in deep learning-based methods for glioma segmentation, including network architectures, training strategies, and evaluation metrics.	The study does not provide a direct comparison of different methods and their performance on different datasets.
10	S. S. Chennu et al. (2021)	"Automated detection of meningioma using convolutional neural networks on multi-sequence MRI"	The study proposes a convolutional neural network-based method for automated detection of meningioma using multi-sequence MRI. The proposed method includes pre-processing, feature extraction, and classification.	The proposed method achieves high accuracy in meningioma detection and reduces the workload of radiologists.	The study uses a small dataset and does not compare the proposed method with other state-of-the-art methods.
11	Y. Zhou et al. (2021)	"Multimodal brain tumor segmentation	The study proposes a cascaded 3D convolutional neural	The study proposes a cascaded 3D convolutional neural	The study does not compare the proposed method with

		using cascaded 3D CNNs with model aggregation and label fusion"	network-based method for multimodal brain tumor segmentation. The proposed method includes cascaded segmentation, model aggregation, and label fusion.	network-based method for multimodal brain tumor segmentation. The proposed method includes cascaded segmentation, model aggregation.	other state-of-the-art methods.
12	H. Qi et al.(2021)	"Glioblastoma multiforme detection using deep learning"	The study reviews recent developments in deep learning-based methods for glioblastoma multiforme detection. The review covers various aspects such as network architectures, training strategies, and evaluation metrics.	The study provides a comprehensive overview of the recent advances in deep learning-based methods for glioblastoma multiforme detection, including network architectures, training strategies, and evaluation metrics.	The study does not provide a direct comparison of different methods and their performance on different datasets.
13	M. Shamsi et al. (2021)	"A review of machine learning techniques for meningioma detection"	reviews recent developments in machine learning-based methods for meningioma detection. The review covers various aspects such as feature extraction, classification algorithms, and evaluation metrics.	The study provides a comprehensive overview of the recent advances in machine learning-based methods for meningioma detection, including feature extraction, classification algorithms, and evaluation metrics.	The study does not provide a direct comparison of different methods and their performance on different datasets.

14	M. Abadi et al. (2021)	"A deep learning-based approach for automatic detection of pituitary adenoma "	a deep learning-based method for automatic detection of pituitary adenoma on MRI. The method involves training a convolutional neural network on a large dataset of MRI images.	The proposed method achieves high accuracy in detecting pituitary adenoma on MRI images. The method is also computationally efficient and can be easily integrated into clinical workflows.	The study does not compare the proposed method with other state-of-the-art methods.
15	S. Z. Azmiet al. (2021)	"A review of machine learning techniques for medulloblastoma detection "	The study reviews recent developments in machine learning-based methods for medulloblastoma detection. The review covers various aspects such as feature extraction, classification algorithms, and evaluation metrics.	The study provides a comprehensive overview of the recent advances in machine learning-based methods for medulloblastoma detection, including feature extraction, classification algorithms, and evaluation metrics. T	The study does not provide a direct comparison of different methods and their performance on different datasets.

CHAPTER 3

REQUIREMENTS

Functional requirements for brain tumor detection using convolution neural networks (CNNs) can include the following:

- Data preprocessing: The input data should be preprocessed to normalize and standardize the images, remove noise, and enhance features. This can include techniques such as image resizing, gray-scale conversion, contrast stretching, and histogram equalization.

- Network architecture: The CNN architecture should be designed to handle the complexity of brain imaging data. This can include convolutional layers, pooling layers, and fully connected layers. The number of layers and their sizes should be optimized based on the specific dataset and problem.

- Training data: Sufficient training data should be provided to the CNN to ensure accurate detection of brain tumors. The training dataset should be diverse and representative of the population being studied.

- Validation data: A validation dataset should be used to test the CNN's performance during training and to optimize hyperparameters.

- Hyperparameters: The CNN's hyperparameters should be optimized to improve performance. This can include learning rate, batch size, number of epochs, and regularization parameters.

- Accuracy: The CNN should be able to accurately detect brain tumors with high sensitivity and specificity. The accuracy of the CNN should be evaluated using a separate test dataset.

- Real-time performance: The CNN should be able to operate in real-time to enable fast and accurate detection of brain tumors in clinical settings

- Interpretability: The CNN should be interpretable to enable clinicians to understand how the network is making its predictions. This can include techniques such as visualization of feature maps, saliency maps, and class activation maps.

- Robustness: The CNN should be robust to variations in image quality and noise to enable reliable detection of brain tumors in diverse populations.

3.2 Non-Functional Requirements

Here are 10 non-functional requirements for the project "BRAIN TUMOR DETECTION":

• Accuracy: The system should be able to provide accurate results with a high level of precision.

• Performance: The system should be able to analyze a large amount of data in a short period.

• Reliability: The system should be reliable and stable to avoid any errors or crashes.

• Scalability: The system should be able to handle a large amount of data and should be scalable as the data grows.

• Security: The system should be designed to ensure the privacy and confidentiality of patient data.

• Usability: The system should be user-friendly and easy to use for doctors and healthcare professionals.

• Compatibility: The system should be compatible with different operating systems and devices.

• Maintainability: The system should be easy to maintain and update to ensure its effectiveness over time.

• Availability: The system should be available 24/7 to ensure access to patient data at any time.

• Ethical considerations: The system should be developed following ethical considerations, such as informed consent and data privacy laws.

3.3 System Requirements

3.3.1 Hardware Requirements

• System : Core i3 processors.

• Hard Disk : 500 GB.

• Ram : 8 GB / 12 GB

• Any desktop / Laptop system with above configuration or higher level

3.3.2 Software Requirements

• Operating system: Windows 8 / 10 (64 bits OS)

• Programming Language: Python 3

• Framework: Anaconda

• Libraries: Scikit-learn

• IDE: Jupyter Notebook

• Libraries : Keras, TensorFlow, Numpy,Matplotlip,SkLearn,Mlxtend

CHAPTER 4

DETAILED DESIGN/ARCHITECTURE

The system architecture for Brain Tumor Detection Using CNN & ML Algorithm consists of several components that work together to achieve the goal of predicting the class of tumor.

4.1 System Architecture

System architecture is the conceptual design that defines the structure and behavior of a system. An architecture description is a formal description of a system, organized in a way that supports. Reasoning about the structural properties of the system. It defines the system components or building blocks and provides a plan from which products can be procured, and systems developed, that will work together to implement the overall system.

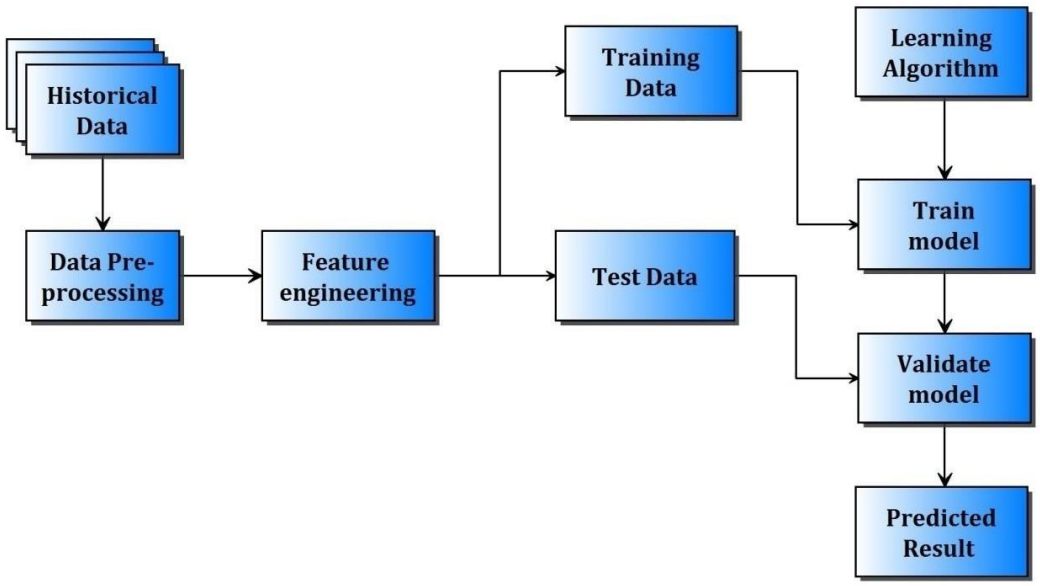


Fig 4.1: System Architecture

4.2 Flow Chart

A flowchart is a picture of the separate steps of a process in sequential order. It is a generic tool that can be adapted for a wide variety of purposes, and can be used to describe various processes, such as a manufacturing process, an administrative or service process, or a project plan.

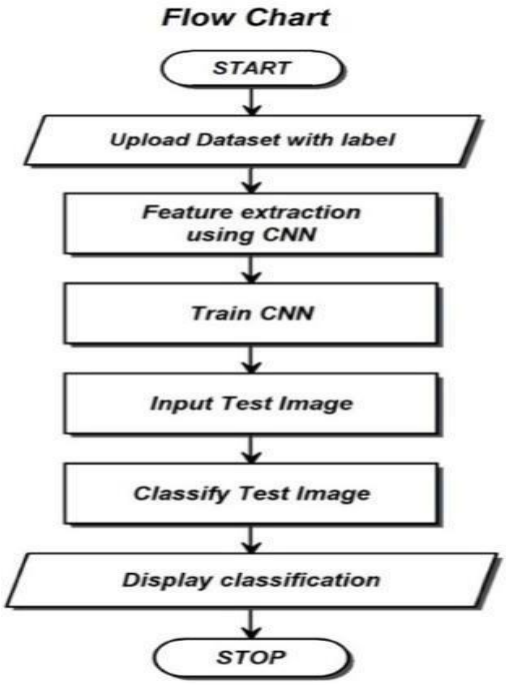


Fig 4.2 Flow Chart

4.3 Use-Case Diagram

A use case diagram is a type of behavioral diagram created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases.

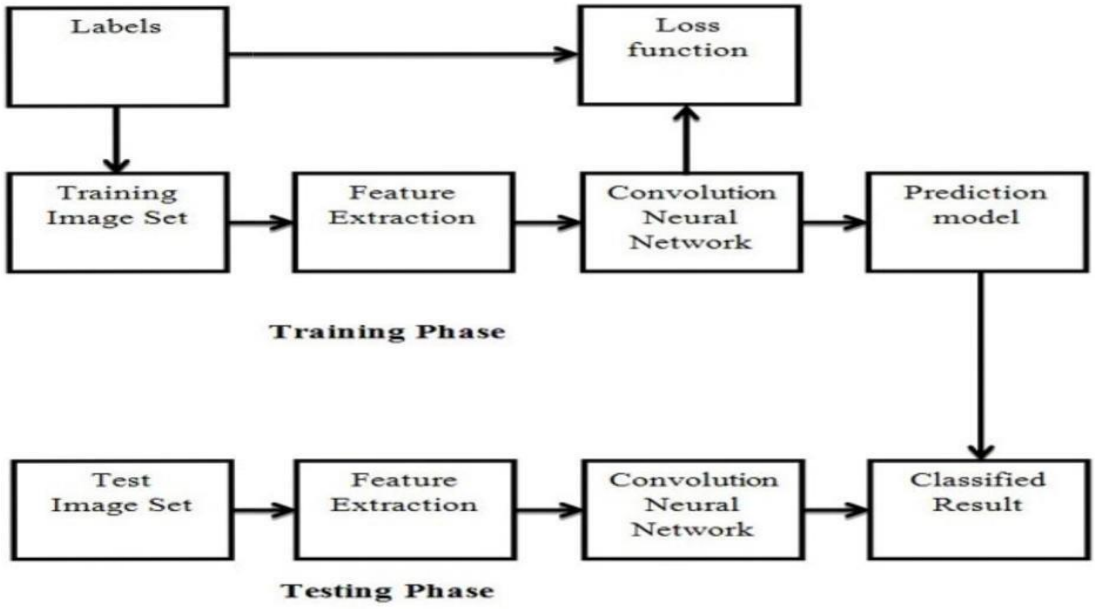


Fig 4.3 Use Case Diagram

4.4 Data Flow Diagram

4.1.1 Level 0 DFD :

A context-level or level 0 data flow diagram shows the interaction between the system and external agents which act as data sources and data sinks. On the context diagram (also known as the level 0 DFD) the system's interactions with the outside world are modeled purely in terms of data flows across the system boundary. The context diagram shows the entire system as a single process, and gives no clues as to its internal organization.

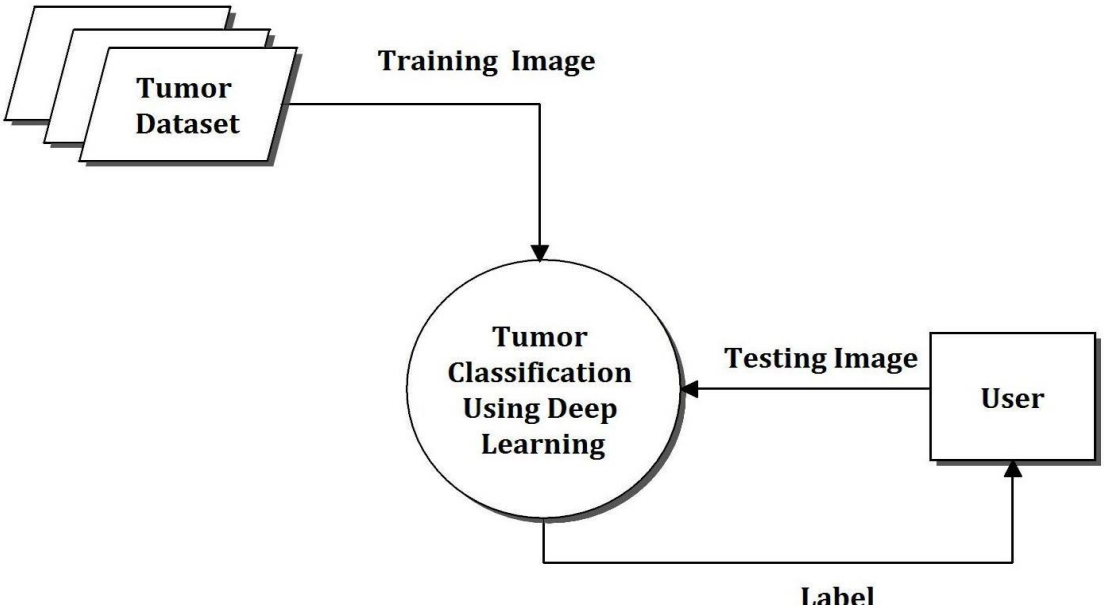


Fig 4.1.1 Level 0 DFD

4.1.2 Level 1 DFD:

The Level 1 DFD shows how the system is divided into sub-systems (processes), each of which deals with one or more of the data flows to or from an external agent, and which together provide all of the functionality of the system as a whole. It also identifies internal data stores that must be present in order for the system to do its job, and shows the flow of data between the various parts of the system.

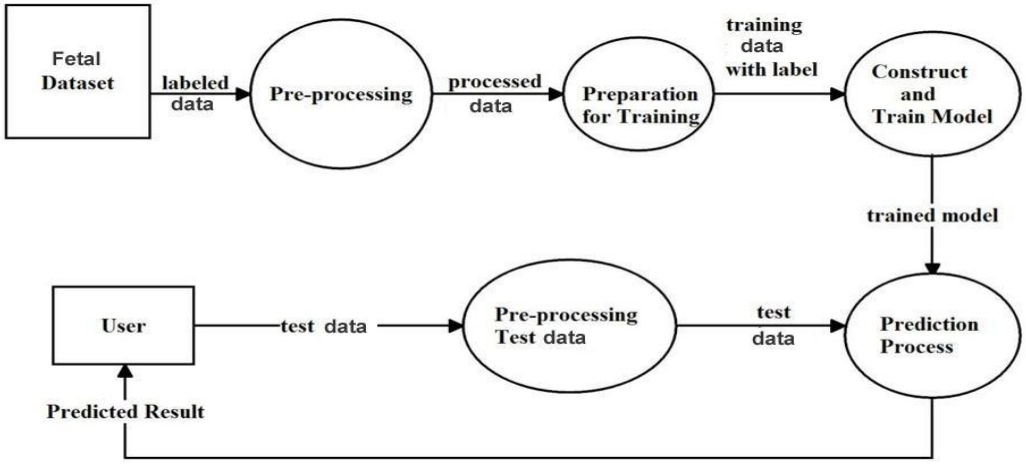


Fig 4.1.2 Level 1 DFD

4.1.3 Level 2 DFD:

The Level 2 DFD goes one step deeper into parts of 1-level DFD. It can be used to plan or record the specific/necessary detail about the system's functioning, Make the process box on the level 1 diagram the system boundary on the level 2 diagram that decomposes it. Identify the processes inside the level 2 system boundary and draw these processes and their data-flows.

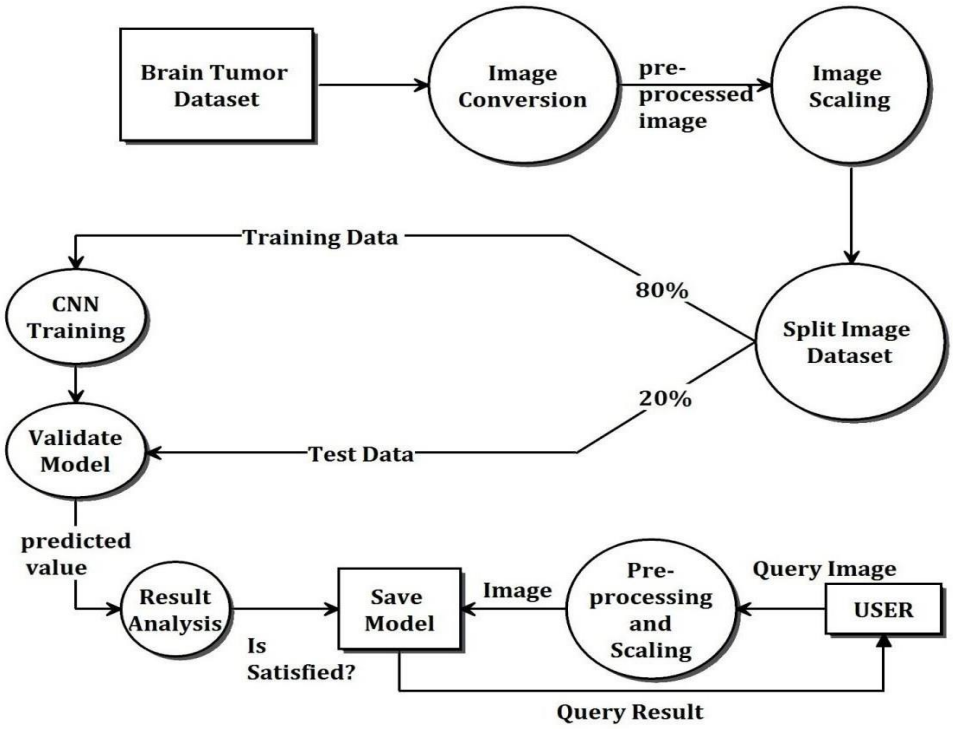


Fig 4.1.3 Level 2 DFD

CHAPTER 5

IMPLEMENTATION

5.1 SDLC (System Development Life Cycle)

It will cover the details explanation of methodology that is being used to make this project complete and working well. Many methodologies or findings from this field mainly generated into journal for others to take advantages and improve as upcoming studies. The method is use to achieve the objective of the project that will accomplish a perfect result. In order to evaluate this project, the methodology based on System Development Life Cycle (SDLC), generally three major step, which is planning, implementing and analysis.

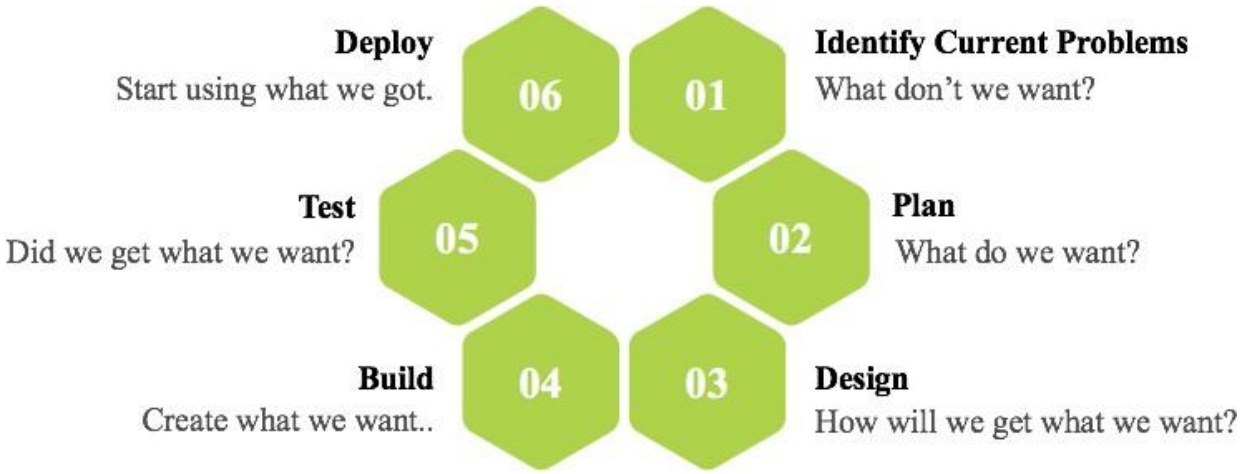


Fig 5.1: Software Development Life Cycle

5.2 Methodology

5.2.1 Planning

5.2.2 Data Collection

5.2.3 Data Pre-processing

5.2.4 Analysis

5.2.1 Planning

To identify all the information and requirement such as hardware and software, planning must be done in the proper manner. The planning phase has two main elements namely data collection and the requirements of hardware and software.

5.2.2 Data collection

Machine learning needs two things to work, data (lots of it) and models. When acquiring the data, be sure to have enough features (aspect of data that can help for a prediction, like the surface of the house to predict its price) populated to train correctly your learning model. In general, the more data you have the better so make to come with enough rows.

The primary data collected from the online sources remains in the raw form of statements, digits and qualitative terms. The raw data contains error, omissions and inconsistencies. It requires corrections after careful scrutinizing the completed questionnaires. The following steps are involved in the processing of primary data. A huge volume of raw data collected through field survey needs to be grouped for similar details of individual responses.

Data Preprocessing is a technique that is used to convert the raw data into a clean data set. In other words, whenever the data is gathered from different sources it is collected in raw format which is not feasible for the analysis.

Therefore, certain steps are executed to convert the data into a small clean data set. This technique is performed before the execution of Iterative Analysis. The set of steps is known as Data Preprocessing. It includes -

- Data Cleaning
- Data Integration
- Data Transformation
- Data Reduction

5.2.3 Data Preprocessing

Data Preprocessing is necessary because of the presence of unformatted real-world data. Mostly real-world data is composed of -

- **Inaccurate data (missing data) -** There are many reasons for missing data such as data is not continuously collected, a mistake in data entry, technical problems with biometrics and much more.

- **The presence of noisy data (erroneous data and outliers) -** The reasons for the existence of noisy data could be a technological problem of gadget that gathers data, a human mistake during data entry and much more.

- **Inconsistent data -** The presence of inconsistencies are due to the reasons such that existence of duplication within data, human data entry, containing mistakes in codes or names, i.e., violation of data constraints and much more.

Structures and a simple but effective approach to object-oriented programming. Python's elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms.

The Python interpreter and the extensive standard library are freely available in source or binary form for all major platforms from the Python Web site, https://www.python.org/, and may be freely distributed. The same site also contains distributions of and pointers to many free third party Python modules, programs and tools, and additional documentation.

The Python interpreter is easily extended with new functions and data types implemented in C or C++ (or other languages callable from C). Python is also suitable as an extension language for customizable applications.

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

- **Python is Interpreted** − Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.

- **Python is Interactive** − you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

- **Python is Object-Oriented** − Python supports Object-Oriented style or technique of programming that encapsulates code within objects.

- **Python is a Beginner's Language** − Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

Python's features include – All these.

- **Easy-to-learn** − Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.

- **Easy-to-read** − Python code is more clearly defined and visible to the eyes.

- **Easy-to-maintain** − Python's source code is fairly easy-to-maintain.

- **A broad standard library** − Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.

- **Interactive Mode** − Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.

- **Portable** − Python can run on a wide variety of hardware platforms and has the same interface on all platforms.

- **Extendable** − you can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.

- **Databases** − Python provides interfaces to all major commercial databases.

- **GUI Programming** − Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.

- **Scalable** − Python provides a better structure and support for large programs than shell scripting.

5.2.4 Analyses

Python community:

One of the reasons for the phenomenal rise of Python is attributed to its ecosystem. As Python extends its reach to the data science community, more and more volunteers are creating data science libraries. This, in turn, has led the way for creating the most modern tools and processing in Python.

The widespread and involved community promotes easy access for aspirants who want to find solutions to their coding problems. Whatever queries you need, it is a click or a Google search away. Enthusiasts can also find access to professionals on Code mentor and Stack Overflow to find the right answers for their queries.

Jupyter Notebook (IDE):

The Jupyter Notebook is an **interactive computing environment** that enables users to author notebook documents that include: - Live code - Interactive widgets - Plots - Narrative text - Equations - Images – Video.

These documents provide a complete and self-contained record of a computation that can be converted to various formats and shared with others using email, Dropbox, version control systems (like git/GitHub) or nbviewer.jupyter.org.

Components:

The Jupyter Notebook combines three components:

- **The notebook web application**: An interactive web application for writing and running code interactively and authoring notebook documents.
- **Kernels**: Separate processes started by the notebook web application that runs users' code in a given language and returns output back to the notebook web application. The kernel also handles things like computations for interactive widgets, tab completion and introspection.
- **Notebook documents**: Self-contained documents that contain a representation of all content visible in the notebook web application, including inputs and outputs of the computations, narrative text, equations, images, and rich media representations of objects. Each notebook document has its own kernel.

Notebook web application:

The notebook web application enables users to:

- **Edit code in the browser**, with automatic syntax highlighting, indentation, and tab completion/introspection.
- **Run code from the browser**, with the results of computations attached to the code which generated them.
- See the results of computations with **rich media representations**, such as HTML, LaTeX, PNG, SVG, PDF, etc.
- Create and use **interactive JavaScript widgets**, which bind interactive user interface controls and visualizations to reactive kernel side computations.
- Author **narrative text** using the Markdown markup language.
- Include mathematical equations using **LaTeX syntax in Markdown**, which are rendered in-browser by MathJax.

Kernels:

Through Jupyter's kernel and messaging architecture, the Notebook allows code to be run in a range of different programming languages. For each notebook document that a user opens, the web application starts a kernel that runs the code for that notebook. Each kernel is capable of running code in a single programming language.

Notebook documents

Notebook documents contain the inputs and outputs of an interactive session as well as narrative text that accompany the code but are not meant for execution. Rich output generated by running code, including HTML, images, video, and plots, is embedded in the notebook, which makes it a complete and self-contained record of a computation.

When you run the notebook web application on your computer, notebook documents are just files on your local filesystem with a ``.ipynb`` extension. This allows you to use familiar workflows for organizing your notebooks into folders and sharing them with others.

Notebooks consist of a linear sequence of cells. There are four basic cell types:

- **Code cells:** Input and output of live code that is run in the kernel
- **Markdown cells:** Narrative text with embedded LaTeX equations

- **Heading cells:** 6 levels of hierarchical organization and formatting
- **Raw cells:** Unformatted text that is included, without modification, when notebooks are converted to different formats using nbconvert.

Internally, notebook documents are `JSON <https://en.wikipedia.org/wiki/JSON>`__data with binary values `base64 <http://en.wikipedia.org/wiki/Base64>`__encoded. This allows them to be read and manipulated programmatically by any programming language. Because JSON is a text format, notebook documents are version control friendly.

Notebooks can be exported to different static formats including HTML, reStructeredText, LaTeX, PDF, and slide shows (reveal.js) using Jupyter's unconvert utility.

Furthermore, any notebook document available from a public URL on or GitHub can be shared via nbviewer. This service loads the notebook document from the URL and renders it as a static web page. The resulting web page may thus be shared with others without their needing to install the Jupyter Notebook.

CHAPTER 6

TESTING

For going for walks the brain tumour detection machine, we utilized the Anaconda platform. Anaconda is an open-source distribution of Python and R programming languages for scientific computing and data science. It affords an effective and convenient environment for developing and going for walks and information-in-depth programs, which include device-gaining knowledge of and deep studying fashions. Via using Anaconda, we had been able to without difficulty manipulate and deploy the desired Python applications and libraries, which include TensorFlow, Keras, and OpenCV, which are vital for imposing the mind tumour detection machine. Moreover, Anaconda presents a consumer-friendly graphical interface and powerful command-line gear for dealing with our Python environment and running our machine, making it an ideal platform for our undertaking.



Fig 6.1: Testing Files

6.1 CNN

6.1.1 Confusion Matrix

A table called a confusion matrix is used to evaluate how well algorithms perform, especially in supervised learning. It displays the frequency with which a model's predictions match the actual values correctly or incorrectly. This analysis can be used to evaluate how well a classification model performs on a database of images. The effectiveness of a model is frequently assessed using precision, recall, and other performance metrics. A two-category classification problem is frequently evaluated using the confusion matrix, which is a crucial source of performance measurements for classification problems.

Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

$$\text{True Positive Rate} = \frac{TP}{TP + FN}$$

$$\text{True Negative Rate} = \frac{TN}{TN + FP}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Fig 6.1.1: Confusion matrix for CNN and formulas

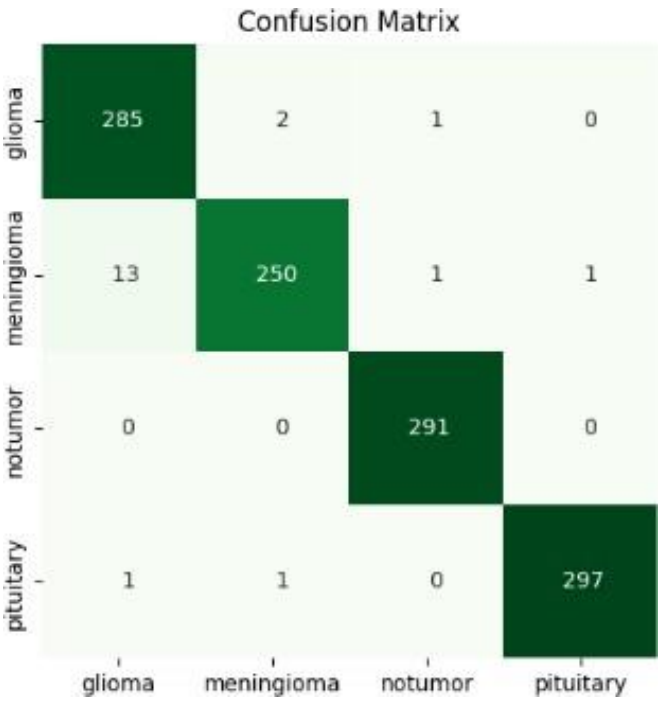


Fig 6.1.2: Confusion Matrix for CLSTM

A confusion matrix aids in visualizing the results of a classification task by providing a table arrangement of the different scenarios of the prediction and findings. It given below fig 21. It creates a table with all of a classifier's predicted and actual values. Fig 14 shows the chart for the confusion CNN model. From the confusion matrix understand this model is performing excellent. Four thousand photos are provided in total and this model got 98% accuracy. 1143 photos are provided in total. And this model correctly predicted 1123 images and only 20 images predicted wrongly, this model got 98% accuracy.

6.1.2 Classification for CNN

	precision	recall	f1-score	support
glioma	0.95	0.99	0.97	288
meningioma	0.99	0.94	0.97	265
notumor	0.99	1.00	1.00	291
pituitary	1.00	0.99	0.99	299
accuracy			0.98	1143
macro avg	0.98	0.98	0.98	1143
weighted avg	0.98	0.98	0.98	1143

Fig 6.1.3: Classification Report for CNN

In the presented table, the precision, recall, and F1-score are reported for four types of brain tumours - glioma, meningioma, nontumor, and pituitary, using a CNN classification model. The model's accuracy is reported to be 0.98, indicating that the model correctly classified 98% of the brain tumor cases. The precision for all tumor types is high, ranging from 0.95 to 1.00, which means that the model made very few false-positive predictions. The recall for all tumor types is also high, ranging from 0.94 to 1.00, indicating that the model identified almost all cases of each tumor type. The F1-score, which considers both precision and recall, is also high for all tumor types, ranging from 0.97 to 1.00. Overall, the results indicate that the CNN model is highly accurate and precise in classifying different types of brain tumours.

CHAPTER 7

RESULTS

7.1 CONVOLUTION NEURAL NETWORK(CNN)

7.1.1 CLSTM - Report on Classification

A classification report is in fact a performance review metric in machine learning. It is used to show the accuracy, memory, F1 Measure, and support of the trained classification model. For the sake of clarity, Fig. 7.1 provides each of the indicators from the CNN classification report.

Metrics	Definition
Precision	Precision is defined as the ratio of positive examples to the sum of such actual and false positives.
Recall	Recall is defined as the ratio of correct positives to all true negatives and false negatives.
F1 Score	A weighted harmonic average of such recall and precision is known as the F1. The projected capacity for the model is higher the closer the F1 score value is near 1.0.
Support	A amount of instances of a class that truly exist in the dataset constitutes the number of supports. It does not differentiate between kinds; it only improves the performance evaluation process.

	precision	recall	f1-score	support
glioma	0.95	0.99	0.97	288
meningioma	0.99	0.94	0.97	265
notumor	0.99	1.00	1.00	291
pituitary	1.00	0.99	0.99	299
accuracy			0.98	1143
macro avg	0.98	0.98	0.98	1143
weighted avg	0.98	0.98	0.98	1143

Fig 7.1: Confusion Matrix for CLSTM

The table shows the precision, recall, and F1-score for four different types of brain tumours - glioma, meningioma, nontumor, and pituitary. The accuracy of the classification model is reported to be 0.98, indicating that the model correctly classified 98% of the brain tumor cases.

The precision for all tumor types is high, ranging from 0.95 to 1.00, indicating that the model made very few false-positive predictions. The recall for all Tumor types is also high, ranging from 0.94 to 1.00, indicating that the model identified almost all cases of each tumor type. The F1-score, which takes into account both precision and recall, is also high for all tumor types, ranging from 0.97 to 1.00. Overall, the model seems to be highly accurate and precise in classifying the different types of brain tumours.

7.1.2 CNN- Confusion Matrix

A confusion matrix aids in visualizing the results of a classification task by providing a table arrangement of the different scenarios of the prediction and findings .It creates a table with all of a classifier's predicted and actual values.

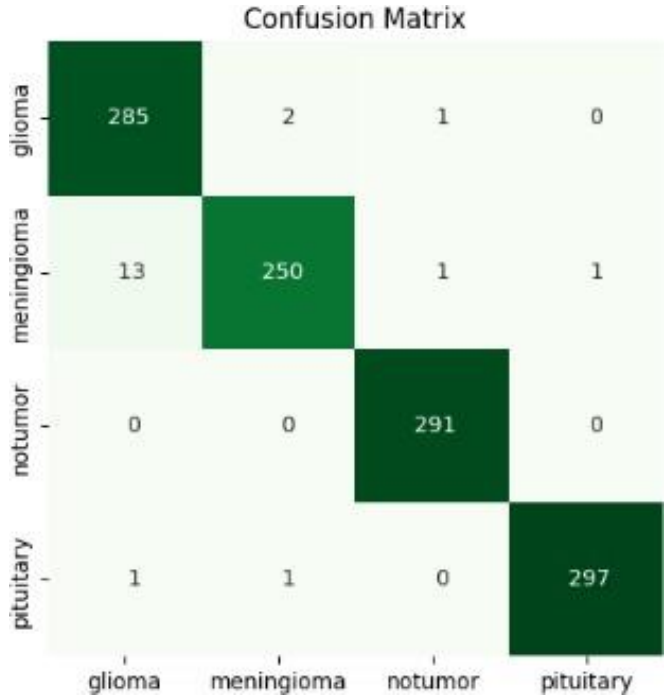


Fig 7.2: Confusion Matrix for CLSTM

Fig 7.2 shows the chart for the confusion CNN model. From the confusion matrix understand this model is performing excellent. Four thousand photos are provided in total and this model got 98% accuracy. 1143 photos are provided in total. And this model correctly predicted 1123 images and only 20 images predicted wrongly, this model got 98% accuracy.

7.2 SNAPSHOTS

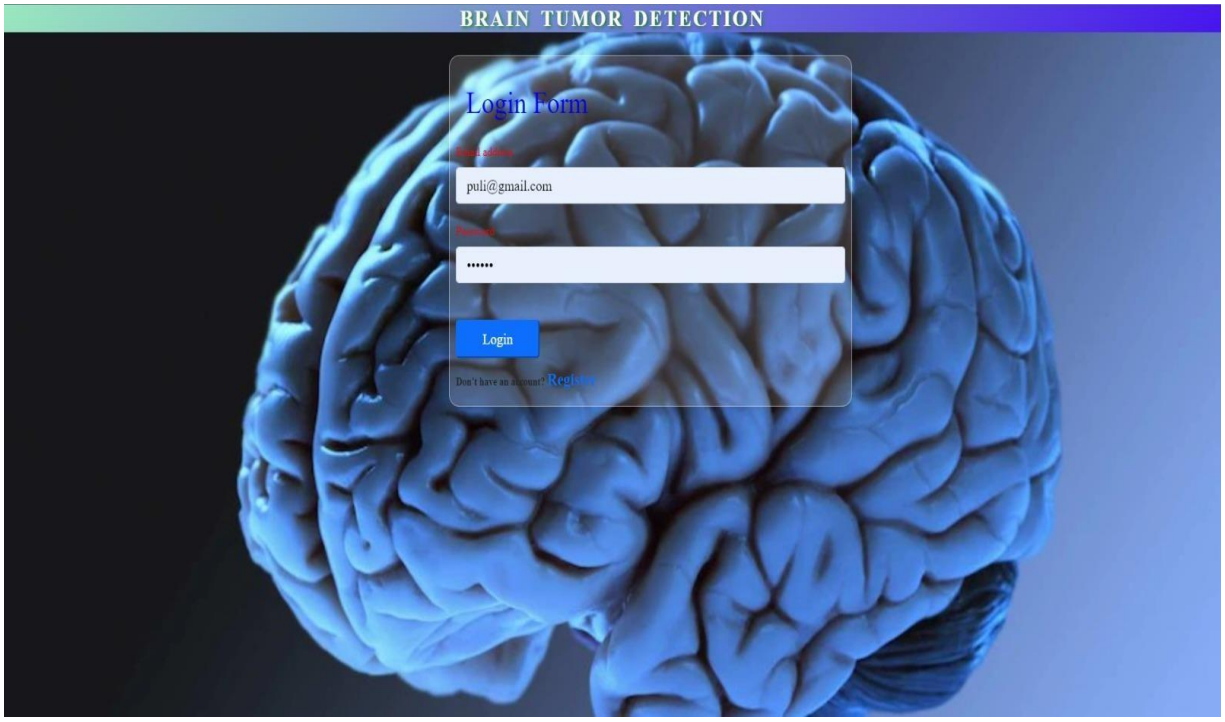


FIG 7.2.1 LOGIN PAGE

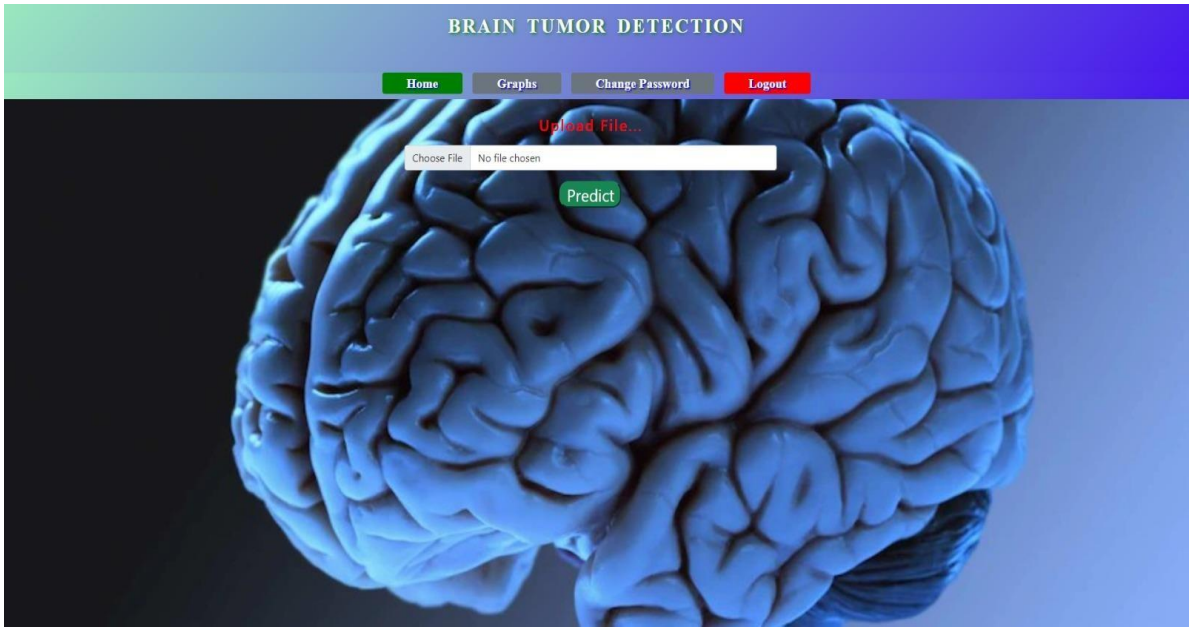


FIG 7.2.2 HOME PAGE

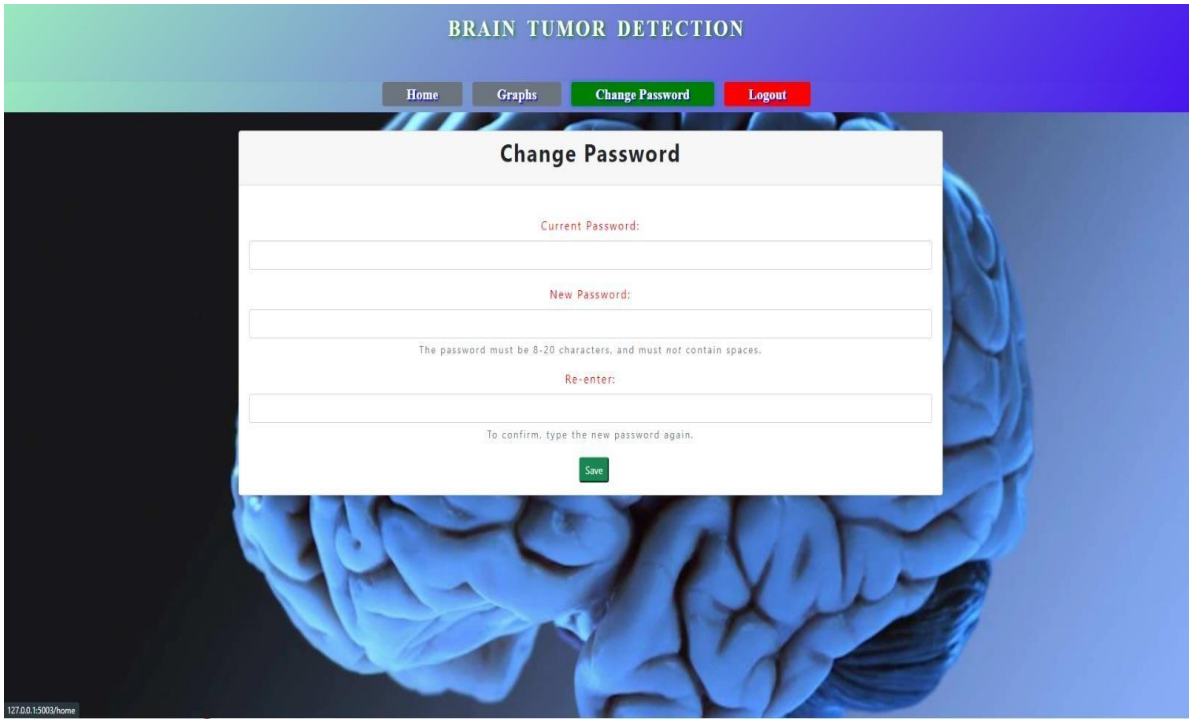


FIG 7.2.3 CHANGE PASSWORD

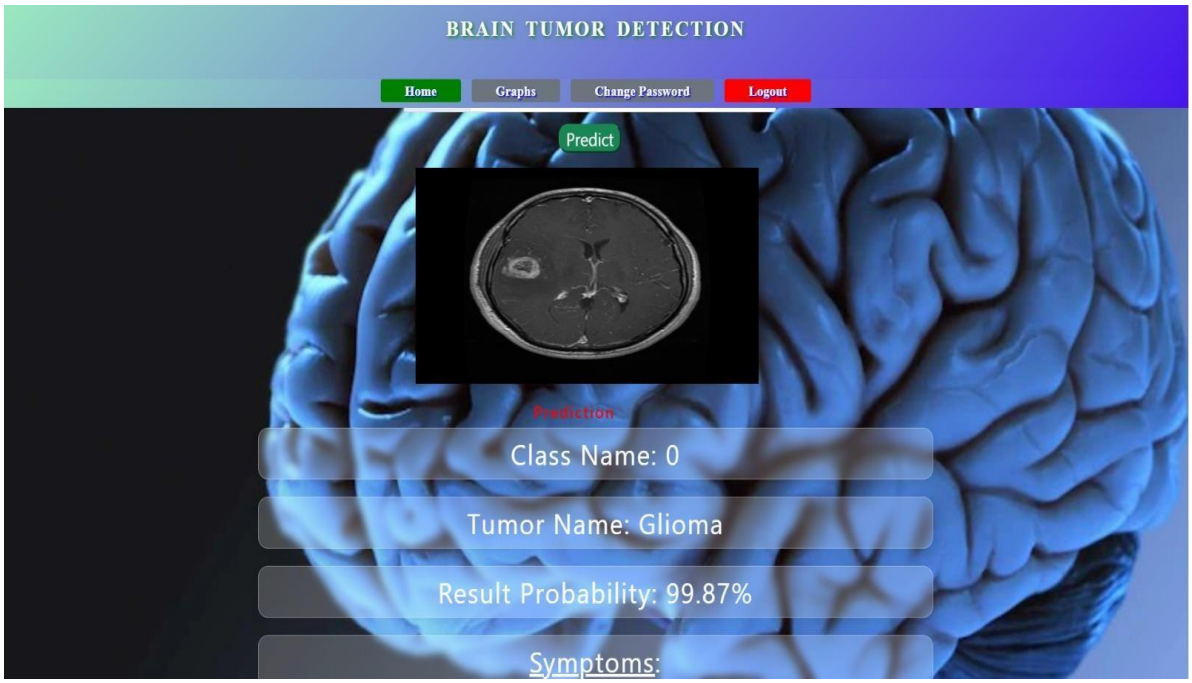


FIG 7.2.4 DETECTION OF TUMOR(GLIOMA)

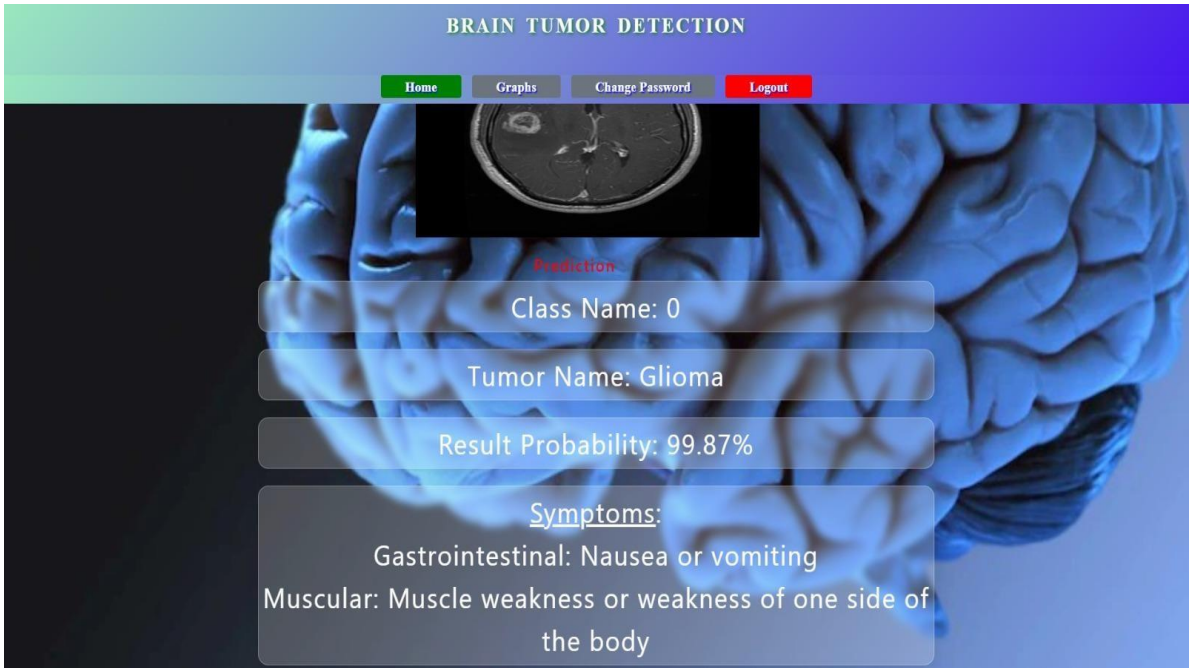


FIG 7.2.5 DETECTION OF TUMOR(GLIOMA)

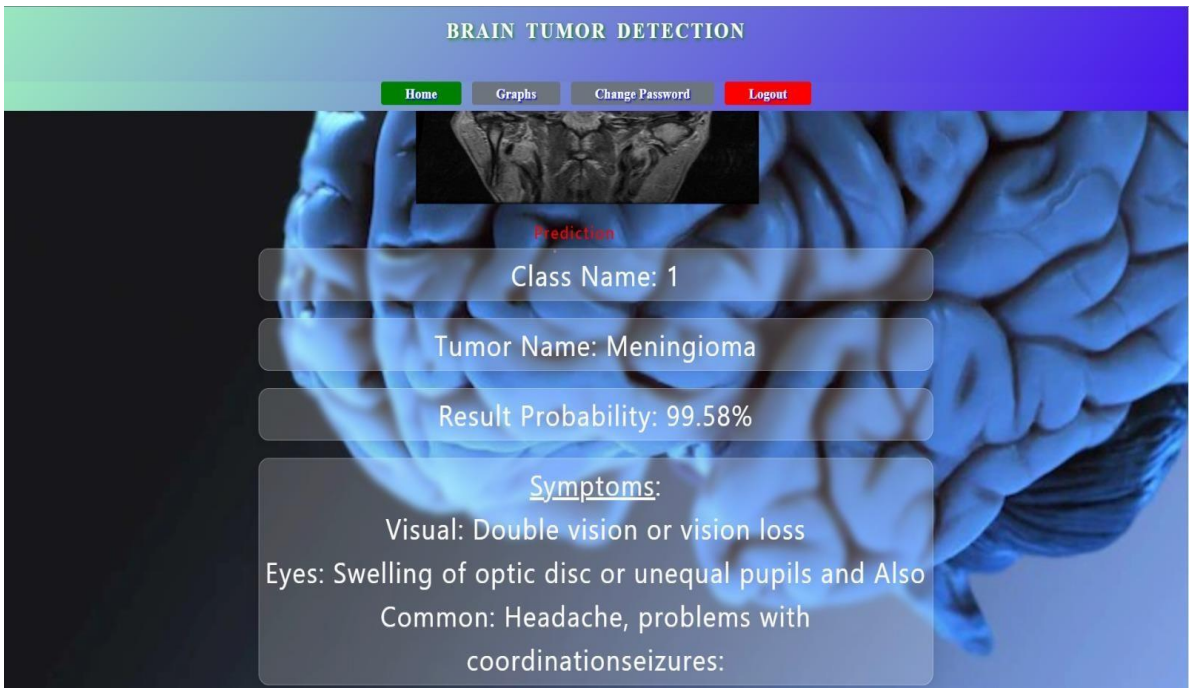


FIG 7.2.6 DETECTION OF TUMOR(MENINGIOMA)

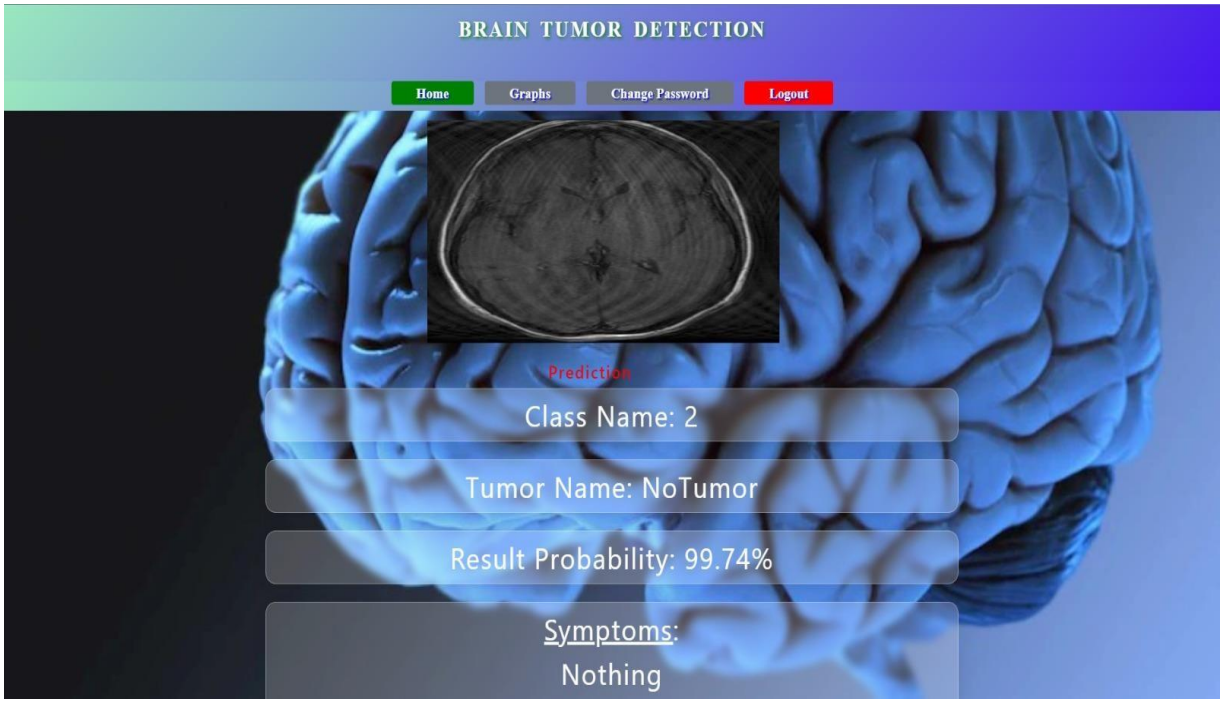


FIG 7.2.7 DETECTION OF TUMOR(NO TUMOR)

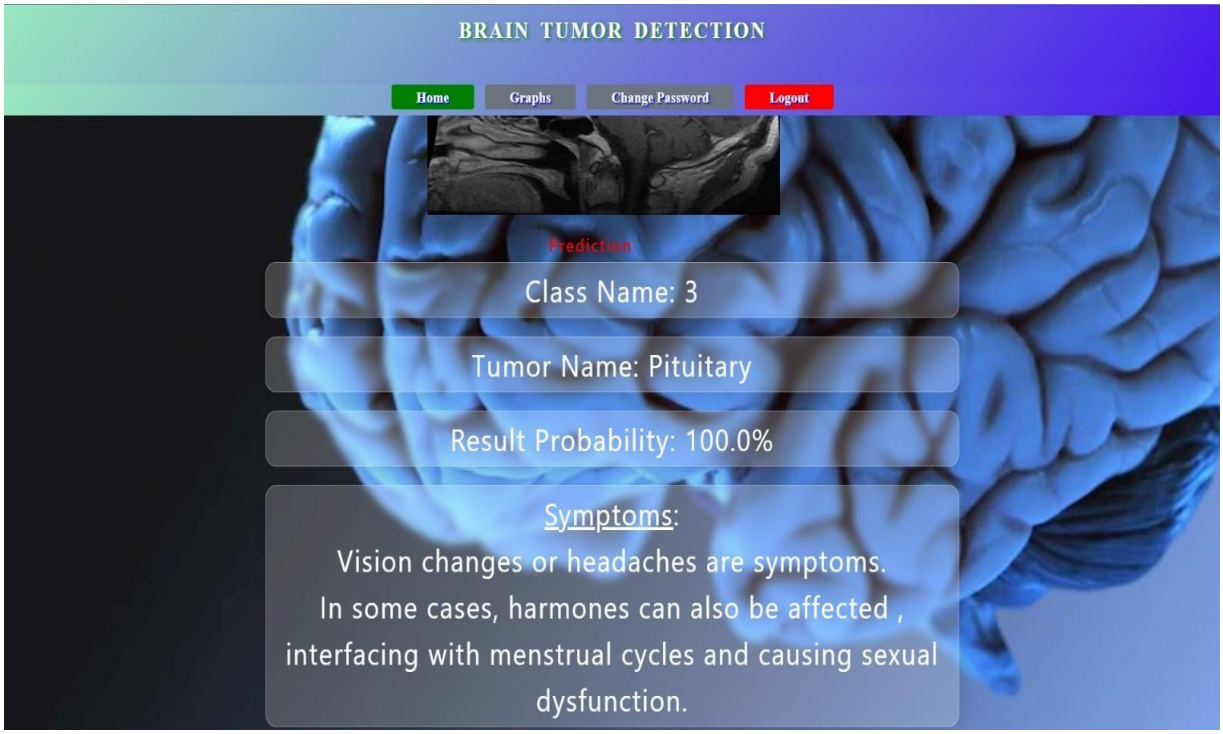


FIG 7.2.8 DETECTION OF TUMOR(PITUITARY)

CHAPTER 8

CONCLUSION

Tumor detection and classification using image processing is a powerful tool in the diagnosis and treatment of cancer. This technique has several advantages over traditional methods, including its non-invasive nature and ability to analyse large amounts of data quickly and accurately.

Image processing algorithms can detect and classify tumours in medical images such as MRI, CT, and PET scans, and provide physicians with valuable information about the size, location, and characteristics of the Tumor. This information can be used to develop personalized treatment plans and improve patient outcomes.

However, there are some limitations to Tumor detection and classification using image processing. These include the need for high-quality images and the potential for false positives or false negatives. Additionally, image processing algorithms may not be able to detect tumours in certain areas of the body or in certain types of cancer.

Overall, tumor detection and classification using image processing is a promising technology that has the potential to revolutionize the way cancer is diagnosed and treated. Ongoing research and development in this area will likely lead to even more accurate and effective tools for detecting and classifying tumours.

8.1 FUTURE SCOPE

- The future scope of tumor detection and classification using image processing is very promising. Advancements in machine learning and artificial intelligence can potentially improve the accuracy and speed of tumor detection and classification.

- In the future, it is possible that image processing algorithms may be able to detect and classify tumors with higher precision, even in areas of the body that are currently difficult to image. This could lead to earlier detection and more effective treatment options for patients.

- Furthermore, the development of imaging technologies such as molecular imaging, which allows for the visualization of molecular events within the body, could potentially provide even more detailed information about tumors and their characteristics. This information could be used to develop even more personalized treatment plans for patients.

- Another area of future development is the integration of image processing algorithms with other diagnostic tools such as genomics and proteomics. By combining data from multiple sources, physicians may be able to better predict a patient's response to different treatments and tailor their therapy accordingly.

- In summary, the future of tumor detection and classification using image processing is very promising. Ongoing research and development in this area will likely lead to even more accurate and effective tools for detecting and classifying tumors, which can improve patient outcomes and ultimately help to reduce the burden of cancer.

REFERENCES

[1] Komal Sharma, Navneet Kaur, " Comparative Analysis of Various Edge Detection Techniques" , International Journal of Advanced Research inC omputer Science and Software Engineering, IJARCSSE, ISSN: 2277 128X, Volume 3, Issue 12, December 2019.

[2] Suchita Goswami, Lalit Kumar P. Bhaiya, " Brain Tumor Detection Using Unsupervised Learning based Neural Network" , IEEE International Conference on Communication Systems and Network Technologies,2020.

[3] Walaa Hussein Ibrahim, Ahmed Abdel Rhman Ahmed Osman, Yusra Ibrahim Mohamed, "MRI Brain Image Classification Using Neural Networks" ,IEEE International Conference On Computing, Electrical and Electronics Engineering, ICCEEE,2018.

[4] Dipali M. Joshi, N. K. Rana, V. M. Misra, " Classification of Brain Cancer Using Artificial Neural Network" , IEEE International Conference on Electronic Computer Technology ,ICECT ,2020.

[5] A. Prabin and J. Veerappan, “Computer Assisted Diagnosis of Brain Tumor in MRI image using Texture features as Input to AdaBoost Classifier”, Research Journal of Applied Sciences, Engineering and Technology 8(24): 2374-2380, 2020

[6] Sahar Ghanavati, Junning Li, Ting Liu, Paul S. Babyn, Wendy Doda, George Lampropoulos, “Automatic Brain Tumor Detection In Magnetic Resonance Images” IEEE 2021

[7] V. Trinadh babu, Sk. Salma Begum, “An Adaptive MRI Tumor Detection Using NeuralNetwork Based Adaboost Algorithm”, International Journal of Computer Science and Information Technologies, Vol. 6 (1) , 2022.

[8] Prachi Gadpayleand, P.S. Mahajani, "Detection and Classification of Brain Tumor in MRI Images ", International Journal of Emerging Trends in Electrical and Electronics, IJETEE – ISSN: 2320-9569, Vol.5, Issue. 1, July-2018.

[9] A Hybrid Deep Learning Framework for Brain Tumor Segmentation and Classification" by Chengliang Dai et al., published in IEEE Access, 2019.

[10] An Improved Brain Tumor Detection and Segmentation Framework Based on Mask R-CNN" by Zhe Li et al., published in IEEE Access, 2020.

[11] A New Automated Brain Tumor Detection and Classification Scheme Based on Convolutional Neural Networks" by Abdelmounaime Lakhssassi et al., published in IEEE Access, 2019.

[12] Brain Tumor Detection and Segmentation Using Convolutional Neural Network and Support Vector Machine" by S. Satapathy et al., published in IEEE Transactions on Computational Imaging, 2019.

[13] Deep Learning-Based Brain Tumor Detection Using MRI Images" by Jingwei Li et al., published in IEEE Access, 2018.

DrillBit

The Report is Generated by DrillBit Plagiarism Detection Software

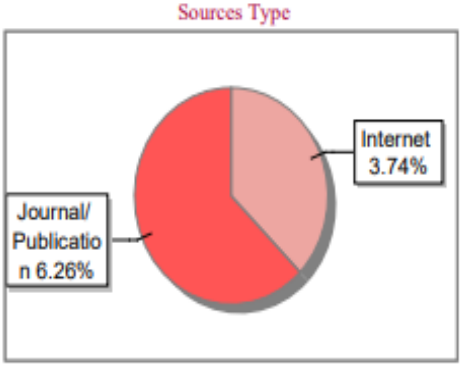
Submission Information

Author Name	Bogathi Anudeepa
Title	Brain tumor detection using CNN
Paper/Submission ID	752039
Submission Date	2023-05-24 14:44:46
Total Pages	40
Document type	Assignment

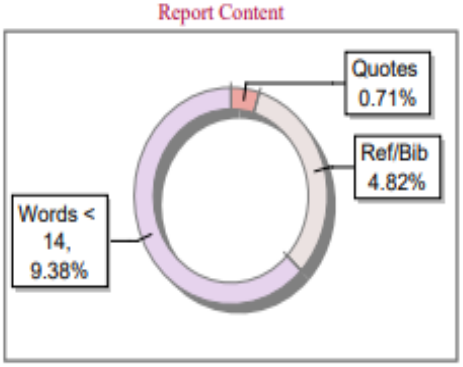
Result Information

Similarity **10 %**

Sources Type



Report Content



Exclude Information

Quotes	Excluded
References/Bibliography	Excluded
Sources: Less than 14 Words Similarity	Excluded
Excluded Source	**0 %**
Excluded Phrases	Not Excluded

A Unique QR Code use to View/Download/Share Pdf File

DrillBit

DrillBit Similarity Report

10	14	A
SIMILARITY %	MATCHED SOURCES	GRADE

A-Satisfactory (0-10%)
B-Upgrade (11-40%)
C-Poor (41-60%)
D-Unacceptable (61-100%)

LOCATION	MATCHED DOMAIN	%	SOURCE TYPE
1	www.tcetmumbai.in	2	Publication
2	www.tcetmumbai.in	1	Publication
3	www.tcetmumbai.in	1	Publication
4	turcomat.org	1	Internet Data
5	moam.info	1	Internet Data
6	mdpi.com	1	Internet Data
7	ACM Press the ACM SIGAda annual international conference- Saint Pe, by Liang, Sheldon X. - 2009	1	Publication
8	www.grahamwheeler.com	1	Internet Data
9	www.ijisme.org	1	Publication
10	A Review on Brain Tumor Diagnosis from MRI Images Practical Implicat, by Khaled Abd-Ellah, M- 2019	<1	Publication
11	studymoose.com	1	Internet Data
12	abnews-wire.blogspot.com	<1	Internet Data
15	arxiv.org	<1	Publication